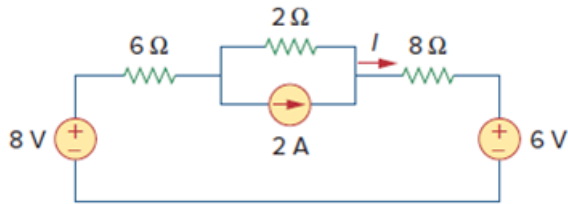


Problem

Find I in the circuit of Fig. 4.14 using the superposition principle.

Figure 4.14:



Step-by-step solution

Step 1 of 7

Draw the circuit diagram as shown in Figure 1.

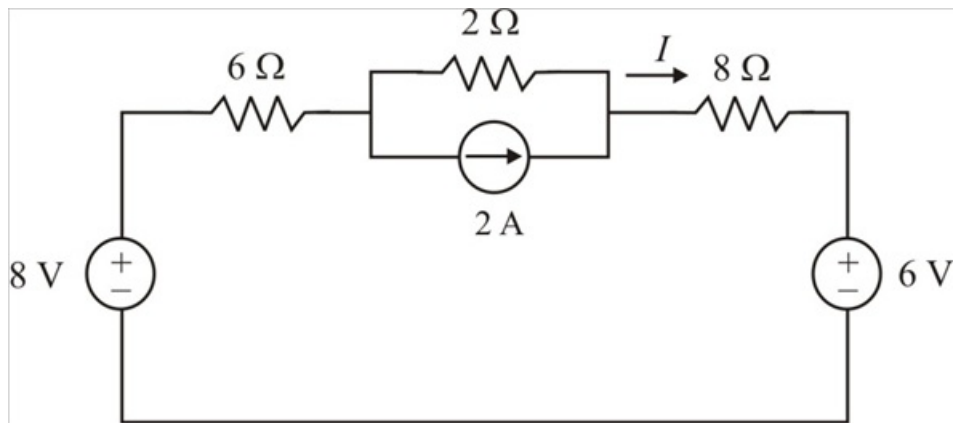


Figure 1

Step 2 of 7

Since there are three sources,

$$\text{Let } I = I_1 + I_2 + I_3$$

Where I_1, I_2 and I_3 are the contributions due to the 8 V, 6V voltage sources and the 2A current source respectively. To obtain I_1 , we set the 6V voltage source and 2A current source to zero as shown in Figure 2.

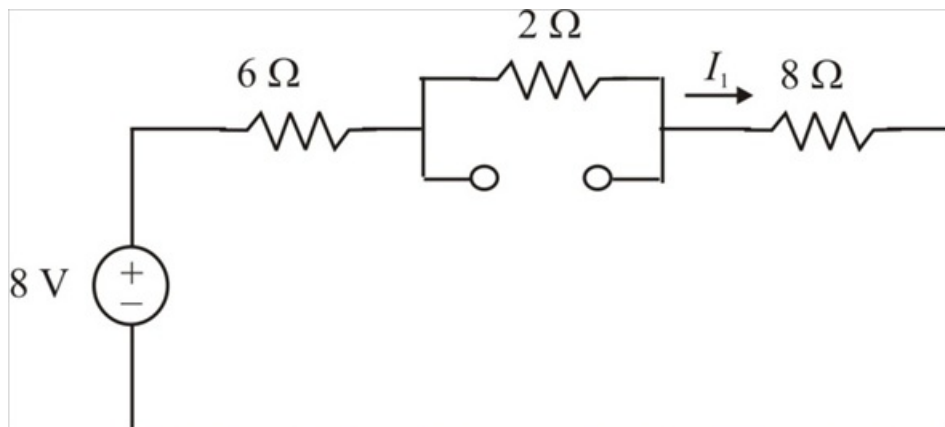


Figure 2

Step 3 of 7

Since the three resistances are connected in series, the current can be calculated as:

$$\begin{aligned}
 I_1 &= \frac{V}{R} \\
 &= \frac{8}{6+2+8} \\
 &= 0.5 \text{ A}
 \end{aligned}$$

Step 4 of 7

To obtain I_2 , we set the 8 V voltage source and 2A current source to zero as shown in Figure 3.

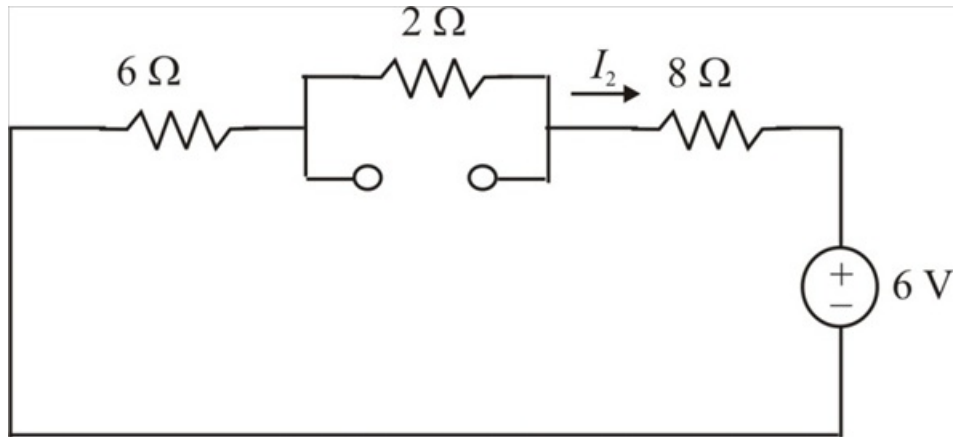


Figure 3

Step 5 of 7

Since the three resistances are connected in series, the current can be calculated as:

$$\begin{aligned}
 I_2 &= -\frac{V}{R} \\
 &= -\frac{6}{6+2+8} \\
 &= -0.375 \text{ A}
 \end{aligned}$$

Step 6 of 7

To obtain I_3 , we set the 8 V and 6 V voltage sources to zero as shown in Figure 4.

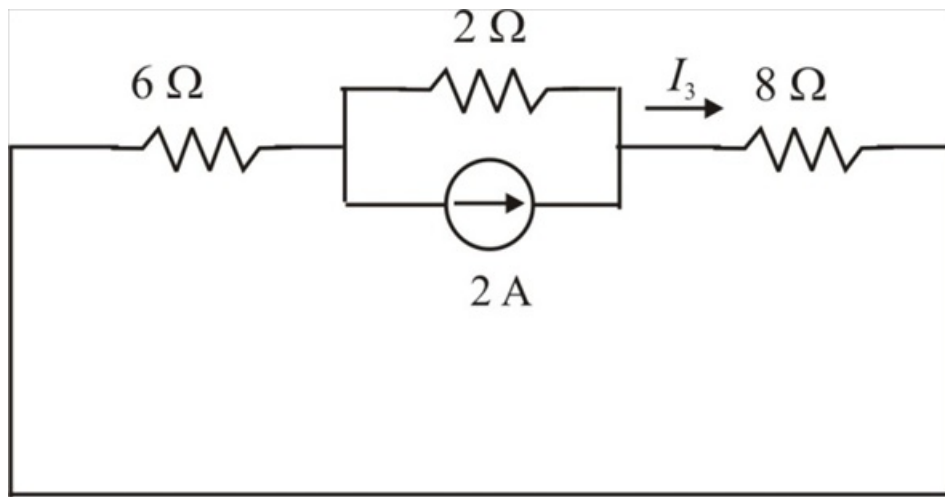


Figure 4

Step 7 of 7

By current division,

$$I_3 = \frac{2}{2+8+6}(2) \\ = 0.25 \text{ A}$$

Thus the current I is:

$$I = I_1 + I_2 + I_3 \\ = 0.5 - 0.375 + 0.25 \\ = 0.375 \text{ A} \\ = 375 \text{ mA}$$

Therefore the current I is 375 mA.